WEB TECHNOLOGY

CREATION OF A FULLY-FUNCTIONAL WEB APPLICATION

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COHORT: BSC (HONS) CYBERSECURITY
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iPHARMATIC BUSINESS OVERVIEW

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Business Overview

iPHARMATIC is a specialized e-commerce platform designed to extend the reach of an established community pharmacy in Mauritius. By transitioning traditional pharmacy services into a secure digital ecosystem, the platform enables customers to browse, search, and purchase both over-the-counter (OTC) and prescription medications remotely. Every prescription order undergoes strict digital verification by licensed pharmacy staff prior to dispatch, ensuring absolute compliance with healthcare regulations.

Mission: To enhance the safety and convenience of pharmacy access for Mauritian consumers through a trusted, verified online service.

Vision: To establish the first digital pharmacy platform in Mauritius, recognized for its operational reliability and rigorous prescription safety standards.

Problem Definition

- **Cumbersome experience**: Consumers face the inconvenience of physical travel or phone-based ordering for every purchase.
- **Manual Handling**: Traditional management workflows lack structured digital tracking, creating documentation gaps.
- **Lack of Records**: A lack of centralized records prevents customers from easily managing medications.
- Difficult Reorders: Customers have no easy way to reorder their recurring medications.
- **Limited Reach**: Existing market alternatives lack a standardized, verified-prescription workflow, leaving a substantial gap for secure online healthcare platforms.

Products or Services

IPHARMATIC provides a comprehensive online catalog featuring over the counter (OTC) and prescription products, along with integrated digital shopping cart and checkout functionalities.

Value Proposition: The platform delivers remote convenience to consumers while strictly maintaining the essential safety protocols and regulatory checks inherent to traditional, in-person pharmaceutical purchases.

Market Analysis

The target market comprises pharmacy consumers within Mauritius who are currently restricted to traditional telephonic or in-person methods.

Primary customer: The primary consumer base consists of both existing and new pharmacy patrons seeking a streamlined, secure online solution for ordering OTC medications and managing recurring prescription renewals.

Competitive Advantage

The primary differentiator for iPHARMATIC lies in its proprietary prescription-verification workflow. Restricted pharmaceutical items cannot be cleared for shipping until a licensed, on-site pharmacist formally reviews and approves the uploaded medical documentation. This rigorous process builds high consumer trust and regulatory compliance—barriers to entry that generic e-commerce platforms and standard delivery services cannot easily replicate.

Operations

Hours	M-F: 8:00 AM- 6:00 PM / Sat: 8:00 AM – 1:00 PM /Sun: 9:00 AM – 12:00 PM
Facility	500 sq ft centralized dispensary, inventory storage area, and dedicated packing dispatch station

Finances

Startup Capital	Rs 350000
Yr 1 Revenue	Rs 1200000
Yr 1 Net Income	Rs 180000
Funding Request	Rs 0

iPHARMATIC – Functional and Security

Actors defined (guest, customer, pharmacist(staff), admin, delivery rider)

Actors are external to the system.

- An actor represents the role of a user that interacts with the system, in our case it is our pharmacy that we are currently modelling. We can represent multiple users with a single actor, and a single user can have the role of multiple actors.

Different types of actors involved:

Actor	Requirement	Category
Guests (Unregistered Visitor)	• They are unauthenticated users to the site • They are not registered in/ logging in yet *Boundary: cannot upload prescriptions and cannot add items to cart as customers*	
	Guests should be able to: • Browse and search the medicinal catalogue even without an account • See the reviews posted by customers and product details • Register for an account or log in to an existing account so that they can make a purchase • Be redirected to the login page when adding, or purchasing any product if they have not created an account yet	Functional
	Guests must have a/an: • Rate limiting on the login page to reduce automated abuse • Monitored search and filter inputs using prepared statements to avoid any SQL injection	Security

Actor	Requirement	Category
Customer (Registered Buyer)	• They are usually the ones who are registered in the web application *Boundary: should not be able to view other customers' prescriptions and payments type.*	
	Customer should be able to: • Add, purchase, track delivery, upload and get prescription approvals, refunds, leave a review, or adjust quantities of products and view their purchasing history • Customers should be able to upload their prescriptions on the website to be reviewed by the pharmacist so that checkouts are allowed • Able to have different types of payment methods	Functional
	Customer should have a/an: • Encrypted login passwords and card credentials using strong **salted hash** (What is a salt in hashing?, 2024) • Verified logged-in sessions to be able to check that orders and prescriptions are theirs to prevent **Insecure Direct Object Reference (IDOR)** (Insecure direct object references (IDOR), n.d.)	Security

Actor	Requirement	Category
Pharmacists (Staffs)	• They verify the prescriptions of the customers, and they need to check the delivery process for their customers. *Boundary: should not view other staff's accounts or do admin-only tasks*	
	Pharmacist should be: • Able to monitor the inventory level of medications to prevent it reaching the low-stock level • The pharmacist should be able to provide advice and respond to the queries asked for by the customers • Allowed to log in and access a dedicated staff dashboard, whereby approval or denial of prescriptions must be possible, to view pending prescription reviews, and to view tracking status (processing → packed → dispatched → delivered)	Functional
	Pharmacist should be: • Allowed to review the uploaded prescriptions by the customers and if falsification is detected they should flag • Allowed to approve/reject any prescription after verifying if it belongs to the actual customer, and it must be logged using the pharmacist ID to be reviewed for future audit trails along with a clear reasoning • Monitored when responding to customers' queries	Security

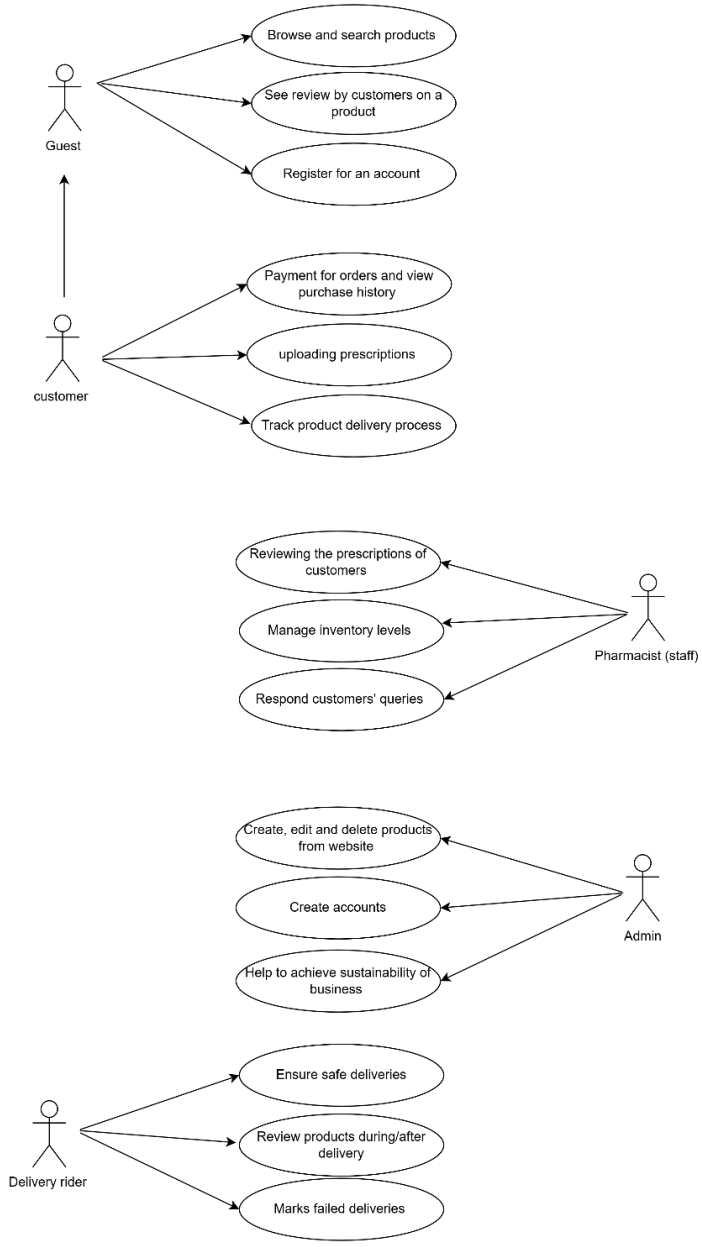
Actor	Requirement	Category
Admin (Manager)	• Responsible for supply chain management, financial management *Boundary: should not perform pharmacists' tasks such as viewing prescriptions of customers*.	
	Admin should be: • Allowed gust and customers privilege, and to access to create, update, read, or delete products from the website • Allowed to view sales, inventory reports, and monitor any inappropriate reviews on any product • Allowed to create new accounts such as customers and pharmacists. • Allowed to manage staff accounts and roles, perform backups, and to view logs of approval or rejected prescriptions	Functional
	Admin should be: • Logged on to every action and decision with their ID to ease audit trails if something goes wrong • Have an included CSRF tokens on all admins CRUD forms (Sharif, 2022) and needs to be validated on server-side to prevent any cross-site scripting	Security

Actor	Requirement	Category
Delivery Rider	• The delivery rider is normally for whoever physically transports the products to customers from the pharmacy. It operates as an external courier service. *Boundary: cannot perform the task of admin and pharmacist*	
	Delivery rider must be: • Allowed to login to a dedicated app, to show lists of orders that need to be delivered, and to display specific customer information for delivery such as delivery address, customer name, and phone number • The delivery rider should ensure that the products are delivered safely without any damage • Allowed to update the delivery status of any order (picked up → out for delivery → delivered) and to take any picture upon delivery as proof • Able to mark failed deliveries	Functional
	Delivery rider must: • Have a restricted limit on only viewing, updating orders, and proof of delivery on the dedicated app • Allow the picture taken to be of a specific file type such as JPG or PNG, and tracking done must be logged with the rider's ID in case of monitoring	Security

Actor	Requirement	Category
All Actors	All actors must: • Be able to view products and reviews comment section	Functional
	All actors must: • Have a prepared statement on all database queries, whereby string concatenation into SQL must not be used • Have all their dynamic content to be output-encoded (Group, 2026) to mitigate cross-site scripting • Have least privilege according to each actor's role to enforce access control on server-side for any request • Have an automatic session termination after a certain amount of time of inactivity to force each user to authenticate themselves • Have a hash of their password using modern encryption algorithm rather than storing them in plain text	Security

USE CASE DIAGRAM

The diagram below shows five actors against the iPHARMATIC system boundary. The line from customer up to guest is **generalization**, generalization is the process of extracting common properties from a set of entities and creating a generalized entity from it (geeksforgeeks, 2026).Hence this means the customers inherits use case connected to guest, such as browsing or search for products and registering or login for an account.



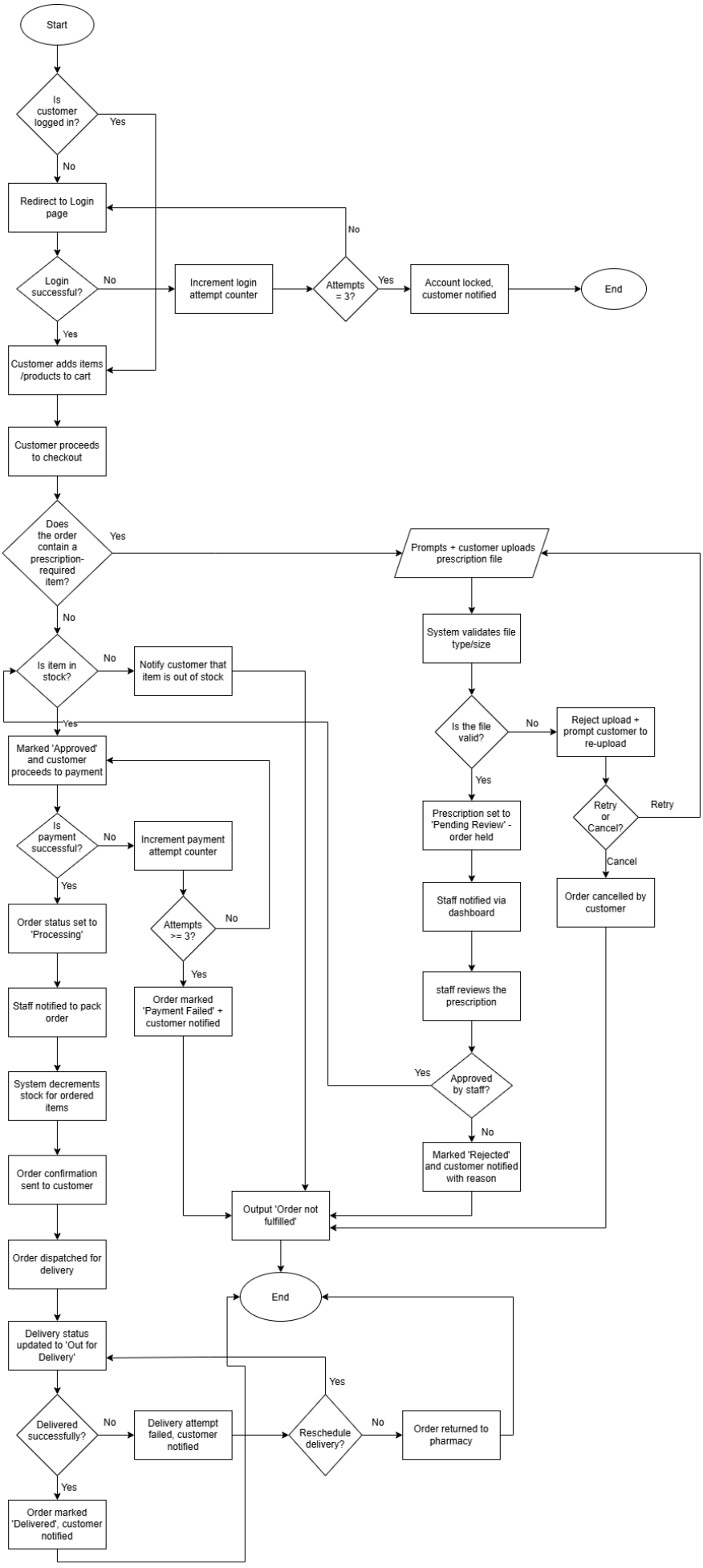
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Use case per actor:

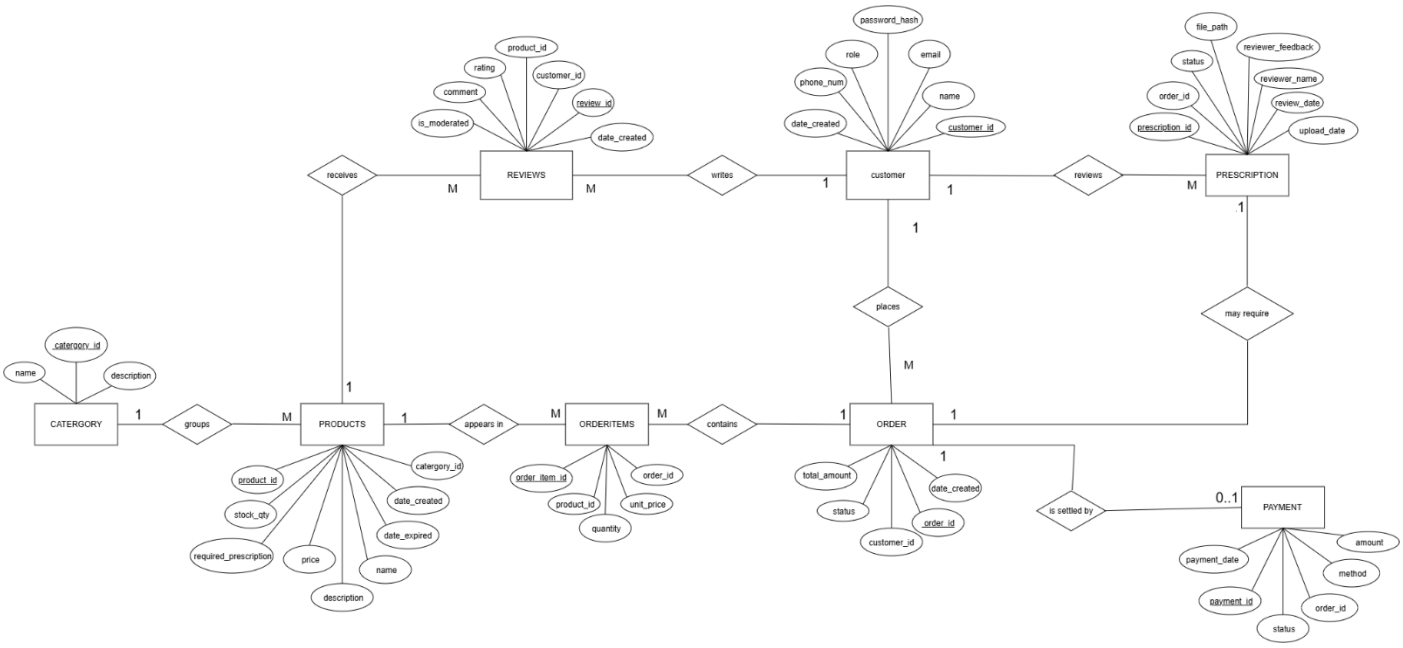
Actor	Use case	Explanation and security relevance
Guest	Browse and search	View products and filter searches. Hence the website should use parameterized queries instead of directly inserting the values from user input as this will protect the system against SQL injection
	Register for an account	While registering for an account, the website should enforce strong passwords rule, store hash passwords and use email verification.
	View products and view reviews in reviews section	If the guest can view the products reviews, then the same parameterized queries should be applied, so that it prevents SQL injection.
Customers	Pay for orders	To ensure the safety of payment, the checkout should be protected using tokenized payment security or implement multi-factor authentication.
	Track product delivery process	Ensuring personal information stays secretive, a strict user authentication before revealing personal details.
	Upload prescriptions	While uploading prescriptions on the website the file type and size should be validated by the server-side. • Server-side: operations are managed behind the scenes on a remote system rather than on the user's device. Hence, this model ensures powerful and secure handling complex tasks (lenovo, n.d.)
Admin	Create, edit and delete products	The business should implement Role based access by granting this following access only to the admin. Hence, it reduces the risk of unauthorized access.
	Create new accounts	The system should use CSRF (cross-site request forgery) protection since it prevents malicious tricks an authenticated user's web browser performing an unwanted action.
	Apply for update, patch and do regular backup	The admin should update the system regularly to include new security features, fix the vulnerabilities that hackers can easily exploit and regular backup need to be done to prevent any data loss.
Pharmacist	Reviewing customers' prescriptions	To ensure the safety of reviewing prescriptions, role-based access control should be implemented so that only pharmacists with this access can review prescriptions.
	Manage inventory levels	Quantity must be validated as a nonnegative integer server-side to prevent negative stock or even overflow.

	Advise and respond to customers	Use encrypted chat as it will ensure that the communication is carried out securely and the customers' information should be protected by GDPR
Delivery rider	Marked failed deliveries	The rider's account must have a restricted limit on only viewing, updating orders and proof of delivery

iPHARMATIC Flowchart



iPHARMATIC ERD



Keys

1 - One
M - Many
0..1 - Zero or one

Relationship Set

Entity Set

Attributes

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